FitBuddy — Phase 2: Requirement Analysis

Functional and non-functional requirements

Purpose

• Prepared from the supplied FitBuddy project document. This document is a submission-ready phase artifact/template; it does not claim that implementation, testing, screenshots, or deployment have been completed unless you add evidence from your actual project.

Functional Requirements

• FR-01: Accept username/user ID, age, weight, fitness goal, and workout intensity.

• FR-02: Generate a personalized 7-day workout plan.

• FR-03: Include warm-up, main workout, sets/repetitions or duration, and cooldown/recovery guidance where generated.

• FR-04: Generate a concise nutrition or recovery tip.

• FR-05: Accept feedback on an existing plan.

• FR-06: Regenerate the plan using the original plan plus user feedback.

• FR-07: Store user information and generated plans in SQLite through SQLAlchemy.

• FR-08: Preserve original and updated plans.

• FR-09: Provide an admin view listing users and their plans.

• FR-10: Provide FastAPI API documentation through /docs.

Non-Functional Requirements

• Usability: the interface should clearly guide users from input to generated results.

• Responsiveness: the supplied design specification calls for mobile responsiveness.

• Maintainability: use a modular FastAPI structure with separate AI, database, routing, and template responsibilities.

• Security: keep the Gemini API key outside source code using an environment variable or .env file.

• Performance: use Gemini Flash for quick nutrition/recovery-tip generation and Gemini Pro for richer plan generation/update according to the supplied project document.

• Persistence: user and plan information should remain available through SQLite storage.

Technology Requirements

• Python.

• FastAPI and Uvicorn.

• Google Generative AI / Gemini API.

• HTML, CSS and Jinja2.

• SQLAlchemy and SQLite.

• python-multipart for form handling.

• Git/GitHub for version control.

Input / Output Summary

• Input: username, user ID, age, weight, goal, intensity.

• Output: personalized 7-day workout plan plus nutrition/recovery tip.

• Feedback input: user ID plus requested changes.

• Updated output: revised workout plan and updated nutrition tip.

• Admin output: user details plus original/updated plans.

Acceptance Checklist

• All required user fields are accepted.

• A plan can be generated and displayed.

• A nutrition/recovery tip is displayed.

• Feedback can update an existing plan.

• Records are stored in SQLite.

• Admin view displays stored records.

• Application can be run locally.